NAME: SWAGATO DHARA

LINKEDIN: www.linkedin.com/in/swagato-dhara-35b706391

PROJECT: STUDY CHATBOT

Project Overview

The Study Bot is an AI-powered educational assistant designed to help students understand complex academic concepts and prepare for their studies. Built using Python and the FastAPI framework, the application operates as a robust RESTful API. It leverages LangChain to orchestrate interactions with a Large Language Model (accessed via the Groq API) and utilizes a custom system prompt to ensure the AI acts as a patient, encouraging, and highly knowledgeable tutor.

To maintain a high-quality learning environment, the bot features built-in guardrails to gently decline non-academic topics and forces structured, scannable formatting for its explanations. The system is fully deployed on Render, allowing users to send queries and receive context-aware responses in real-time.

Explanation of Memory Implementation

To ensure the Study Bot can remember past interactions and provide highly contextual, continuous tutoring, the system implements persistent conversational memory using a MongoDB database.

The memory pipeline functions in four distinct steps for every API request:

1. Data Modeling & Storage: Every interaction is stored as an individual document within a MongoDB collection (named users). Each document records the user_id, the role (whether the message came from the "user" or the "assistant"), the actual message text, and a precise UTC timestamp.
2. Context Retrieval: When a user submits a new question to the /chat endpoint, the application executes the get_history function. This queries the database for all previous documents associated with that specific user_id and sorts them chronologically using the timestamp.
3. Prompt Injection: The retrieved history is mapped into a sequence of conversational turns and injected directly into the LangChain ChatPromptTemplate via a placeholder. This guarantees that the LLM analyzes the entire past conversation before generating its new response, allowing it to refer back to previously discussed concepts.
4. Continuous Persistence: Once the LLM generates its educational response, the backend immediately performs two database insertions: it saves the user's new question, and it saves the AI's newly generated answer. This ensures the database is continually updated and ready for the user's next question.

GITHUB REPO:

https://github.com/swagato022005/study_chatbot.git

HOSTED API LINK:

https://study-chatbot-tkky.onrender.com/docs

EXAMPLES:

```
{
  "user_id": "USER1",
  "question": "MY NAME IS SWAGATO"
}
```

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'https://study-chatbot-tkky.onrender.com/chat' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": "USER1",
  "question": "MY NAME IS SWAGATO"
}'
```

Request URL

```
https://study-chatbot-tkky.onrender.com/chat
```

Response body


```
{
  "response": "Got it, **Swagato**! 😊  \n\nNice to meet you—let's keep the conversation focused on learning. What would you like to explore next?\n\n- **Money-related topics** (e.g., how banks create money, cryptocurrency basics, inflation, personal budgeting)  \n- **Another subject** you're studying (math, science, history, literature, etc.)  \n\nJust let me know, and we'll dive rich t in!"
}
```


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```
{
  "user_id": "USER1",
  "question": "WHAT IS MY NAME AND TELL ME MORE ABOUT MONEY"
}
```

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'https://study-chatbot-tkky.onrender.com/chat' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": "USER1",
  "question": "WHAT IS MY NAME AND TELL ME MORE ABOUT MONEY"
}'
```

Request URL

```
https://study-chatbot-tkky.onrender.com/chat
```

Response body


```
{
  "response": "**Your name is Swagato** – nice to have you here!  \n\nBelow is a deeper look at **money**, building on what we've already covered.\n\n---\n\n## 1. The Modern Monetary System\n
\n| Component | What It Does | Key Points |\n|-----------|--------------|------------|\n| **Central Bank** (e.g., Federal Reserve, ECB, RBI) | Issues base currency, sets policy rates, regulate
s banks | • Controls **interest rates** (policy rate) <br>• Conducts **open-market operations** (buy/sell government securities) <br>• Sets **reserve requirements** for commercial banks |\n| *
*Commercial Banks** | Accept deposits, make loans, provide payment services | • Operate under **fractional-reserve banking** (keep only a fraction of deposits as reserves) <br>• Create **depos
it money** when they lend (money multiplier effect) |\n| **Payment Networks** (Visa, Mastercard, ACH, SWIFT) | Move money electronically between banks and merchants | • Enable **real-time** or
near-real-time transfers <br>• Support **card payments**, online purchases, cross-border remittances |\n| **Government Treasury** | Manages fiscal policy (taxes, spending) | • Issues **governm
ent bonds** that banks may buy/sell, influencing money supply |\n| **Regulators** (FDIC, OCC, European Banking Authority) | Protect depositors, ensure stability | • Provide **deposit insurance
** (e.g., $250k per account in the U.S.) |\n\n---\n\n## 2. Money Creation in Detail\n\n1. **Base Money (High-Powered Money)**\n   - Physical cash + reserves that banks hold at the central ban
k.\n   - Directly controlled by the central bank.\n\n2. **The Money Multiplier**\n   - When a bank receives a deposit, it keeps a **reserve ratio** (e.g., 10 %).  \n   - It can lend out the re
maining 90 %.  \n   - The borrower spends the loan; the recipient deposits it at another bank, which repeats the process.  \n   - **Formula (simplified):**  \n     \\[\n     \\text{Money Multi
plier} = \\frac{1}{\\text{Reserve Ratio}}\n     \\]  \n   - Example: 10 % reserve → multiplier ≈ 10 → a $1,000 deposit can ultimately support up to $10,000 of total deposits.\n\n3. **Quantitat
ive Easing (QE)**\n   - Central banks buy large amounts of government or corporate securities, injecting **additional reserves** into the banking system.  \n   - Goal: lower long-term interest
rates and stimulate lending when conventional policy rates are already near zero.\n\n---\n\n## 3. Types of Money – Expanded\n\n| Type | Characteristics | Typical Uses |\n|------|-------------
--|-------------|\n| **Commodity Money** | Intrinsic value (gold, silver, cattle) | Early economies, store of wealth |\n| **Fiat Money** | No intrinsic value; value comes from government decr
ee | Modern paper/coin currency, most everyday transactions |\n| **Bank Money (Deposits)** | Digital ledger entries; created by banks when they lend | Everyday purchases via debit cards, onlin
e transfers |\n| **Electronic Money (e-money)** | Stored electronically (e.g., PayPal, mobile wallets) | Fast online payments, cross-border remittances |\n| **Cryptocurrencies** | Decentralize
d, blockchain-based, limited supply (e.g., Bitcoin) | Borderless transfers, digital assets, speculative investment |\n| **Stablecoins** | Crypto-backed by fiat or assets to keep price stable
(e.g., USDC) | Bridging fiat and crypto, payments, DeFi platforms |\n\n---\n\n## 4. Inflation & Purchasing Power\n\n- **Inflation** = General rise in price levels → **decrease** in money's pur
chasing power.\n- **Common measures:** Consumer Price Index (CPI), Producer Price Index (PPI), core inflation (excludes food & energy).\n- **Causes of inflation**\n  1. **Demand-pull:** Aggreg
ate demand outpaces supply.  \n  2. **Cost-push:** Rising production costs (wages, raw materials) raise prices.  \n  3. **Monetary:** Excess growth of money supply relative to output.\n\n### Q
uick Formula\n\\[\n\\text{Real Value} = \\frac{\\text{Nominal Value}}{1 + \\text{inflation rate}}\n\\]\n\n*Example:* If you have $1,000 and inflation is 5 %, the real purchasing power is about
$952.\n\n---\n\n## 5. Interest Rates – Why They Matter\n\n- **Nominal vs. Real Rate**  \n  \\[\n  \\text{Real Rate} \\approx \\text{Nominal Rate} - \\text{Inflation Rate}\n  \\]\n- **Policy Ra
te** (e.g., Fed Funds Rate) influences:\n  - **Borrowing costs** for consumers (mortgages, car loans).  \n  - **Business investment** decisions.  \n  - **Currency value** (higher rates often a
ttract foreign capital, strengthening the currency).\n\n---\n\n## 6. Practical Tips for Understanding Money in Your Life\n\n1. **Track Your Cash Flow**  \n   - Use a simple spreadsheet: Income
→ Expenses → Savings.  \n   - Categorize (needs, wants, savings) to see where money "moves."\n\n2. **Calculate the Cost of Borrowing**  \n   - Use the **APR** (Annual Percentag
re loan offers.  \n   - Include fees and compounding frequency.\n\n3. **Watch Inflation**  \n   - Follow CPI releases (usually monthly).  \n   - Adjust long-term savings goals
```


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